interface on another, and so on. So now each piece is simpler to build.

We've talked to a whole bunch of companies. Some want to get a speciality in network chiplets, or memory chiplets. We want to specialise in AI processor chiplets.

If a company wishes to build a product, they can buy some chiplets as finished pieces and only customise the part that they need to, rather than having to buy all the pieces, as it's just too hard and expensive. Chiplets can be used to break one big problem into smaller problems, and smaller companies can independently solve those problems.

Are there any limitations when it comes to power with so many chips running?

Power is a big problem while designing your chip. There's a power management system. We watch how much power it consumes and how hot it is, and we turn the clock and the voltage up and down, to get the best result. We've picked a good size.

As I said, your brain is made up of clusters of cells. If the clusters are too big, talking to the whole thing is too hard; if they're too small, you have to talk to other things too often.

There's a trade-off about how big the AI processor should be so that you balance the power of computation versus networking. We manage it, actively. We know exactly how much power the chip is using.

So, if someone is building a big rack of computers, there is a cut-off on the amount of power utilised. The power in each box is limited and the temperature is limited. We have to look at all these inputs, and then control computation.

What software programming tools are available for developers to leverage an AI processor's capabilities?

The common languages to write to AI models are PyTorch, TensorFlow, Onyx, and JAX. We support those. We use a program called TBM, that helps a programmer interface to our hardware. Underneath that we wrote our own software stack. That's what we've demonstrated to a small number of customers and we plan to open source our software as well.

What is the difference between processors built using RISC-V architecture and those built by ARM or Intel?

RISC-V is an open source, open standard, which means that if somebody wants to build their own RISC-V processor, they can just do it. ARM, Intel, AMD, etc are proprietary architectures. They control who makes them. There is a coordination body, RISC-V International, that facilitates meetings and conferences so that people can collaborate.

We used RISC-V architecture in each AI processor, and we had to modify it so that it would perform. We added some instructions and some ways so that the processors could talk to each other really fast, and ARM or Intel wouldn't do that. We couldn't build our product without RISC-V.

We're also building a high-performance RISC-V processor that's suitable for servers. We've talked to a number of customers who would like to licence it out from us, and then make their own changes. In a world where innovation is really important, you should be able to build the product you want and make the modifications you need without somebody saying no, and RISC-V allows us to do that.

How is debugging impacted when using open source architecture?

For proprietary software, only the company who owns it gets to see the secrets. I worked on some projects where I had no access to the proprietary software. I had to send a bug report to the company.

Open source is the other way around. There may be a better expert than me, but if something doesn't work if I am using open source, I can see everything and then I can figure out how to fix it myself. While using proprietary software, they'll fix it and they get around to it. With open source, debugging is way better.

What are the challenges of using processors in AI workloads, and how do you address them?

The biggest challenge in AI is things are changing really fast. Two years ago, we got inference and training; then there are language models versus image models and big models versus little models. By big, I mean really big, like thousands of processors!

They've invented something called generative models, in which a user does a lookup, and then uses that result to do multiple lookups. Each one of these techniques has created a whole bunch of new software. You have to make sure your computer is very programmable and is able to handle a diverse set of inputs.

Also, there's a very big range of the size of memory capacity people want. There is a big range of power budgets that people have. So the AI competition is changing very fast, and then the range that it operates on is very high.

What are the future developments and advancements that you see in AI processors?

AI processors will continue to get bigger. We're going to continue to optimise how the mathematics operations work for more efficiency. We're going to make them easier to program, so more people can participate and build high-performance AI models.

We're going to see people building AI into more products. People have talked to us about licensing our AI and RISC-V technology, so they can go off and make their own product that does what they want.

A big chip manufacturer may sell a standard AI card that's quite good and very expensive, but they won't let you change it. We think that flexibility is really important. EFY